

Lab Exercise 14.01: Active Directory Domain Services and Domain Connectivity

IPCONFIG

Windows 10 VM:

```
C:\Users\potato>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 192.168.1.11
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1
```

Windows Defender Firewall with Advanced Security

FileActionViewHelp

Windows Defender Firewall

Inbound Rules

Outbound Rules

Connection Security Rules

Monitoring

New Inbound Rule Wizard

Rule Type

Select the type of firewall rule to create.

Steps:

Rule Type

Program

Action

Profile

Name

What type of rule would you like to create?

☒ Program

Rule that controls connections for a program.

☐ Port

Rule that controls connections for a TCP or UDP port.

☐ Predefined:

AllJoyn Router

Rule that controls connections for a Windows experience.

☐ Custom

Custom rule.

< Back

Next >

Cancel

Core Networking - Multicast Listener Discovery

Core Networking

All

Yes

Allow

Core Networking - Multicast Listener Outgoing

Core Networking

All

Yes

Allow

1:06 AM

3/14/2026

Inbound Rules					
Name	Group	Profile	Enabled	Action	
Ping Allowed		All	Yes	Allow	

```
C:\Users\potato>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

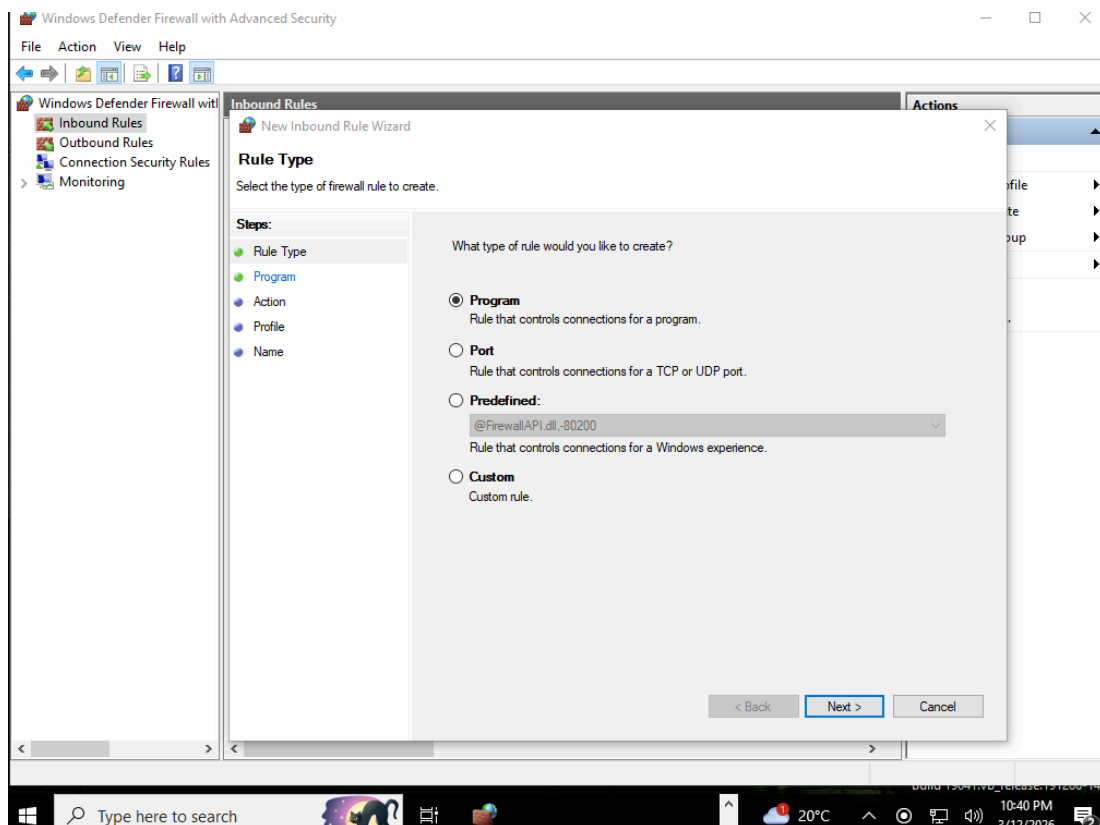
Windows Server 2019 VM

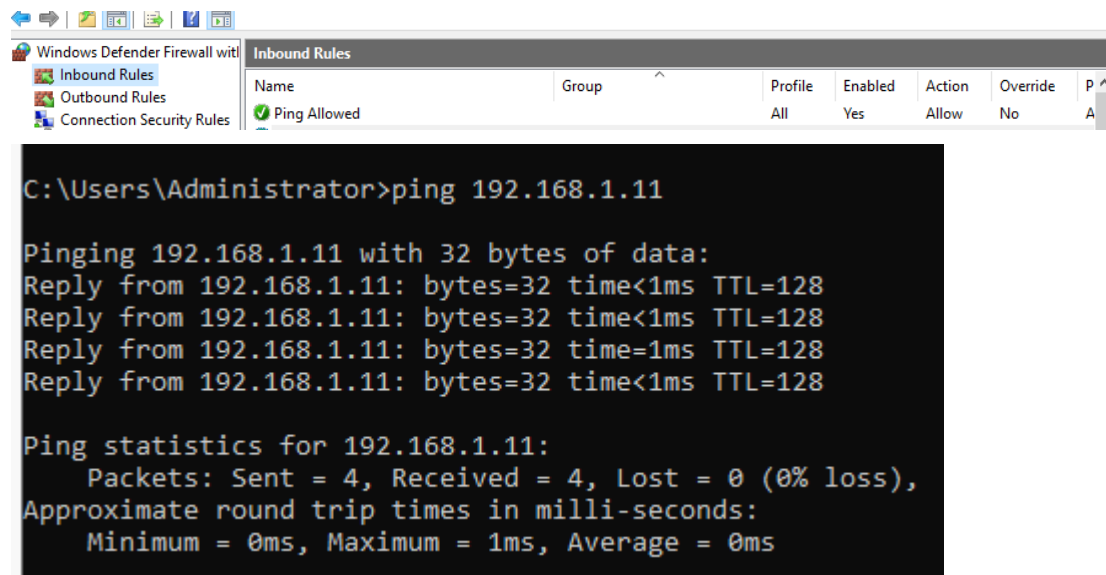
```
C:\Users\Administrator>ipconfig

Windows IP Configuration

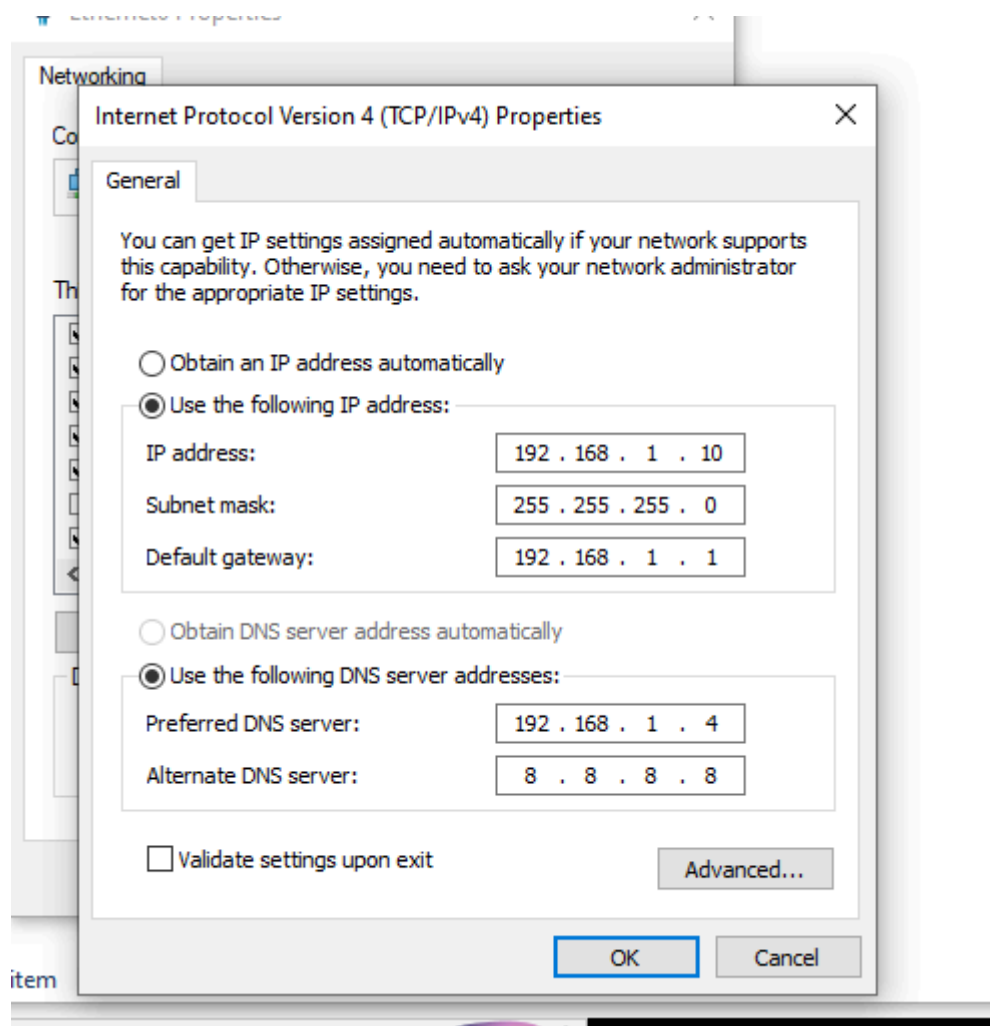
Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2001:ee0:8205:5ddb:f2ad:301b:c0a8:9bbf
    Link-local IPv6 Address . . . . . : fe80::bb0f:f3b9:add9:5e18%5
    IPv4 Address. . . . . : 192.168.1.4
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1%5
                                192.168.1.1
```





Win10 VM IPv4 Ethernet 0 Properties:



[View basic information about your computer](#)

Windows edition

Windows Server 2019 Standard Evaluation

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Windows Server® 2019

System

Processor: 13th Gen Intel(R) Core(TM) i5-13420H 2.61 GHz (2 processors)
Installed memory (RAM): 2.00 GB
System type: 64-bit Operating System, x64-based processor
Pen and Touch: No Pen or Touch Input is available for this Display

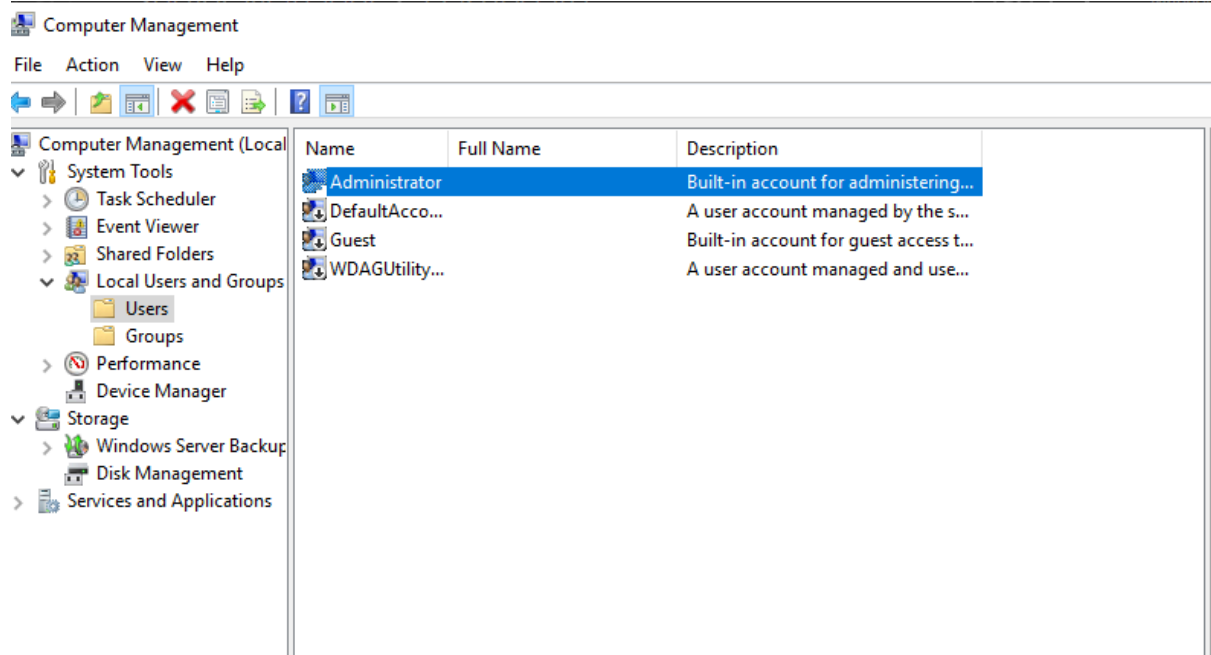
Computer name, domain, and workgroup settings

Computer name: WIN2019-SERVER [Change settings](#)
Full computer name: WIN2019-SERVER
Computer description:
Workgroup: WORKGROUP

Windows activation

Windows is activated [Read the Microsoft Software License Terms](#)

Product ID: 00431-10000-00000-AA944 [Change product key](#)



3a-3j

Add Roles and Features Wizard

— □ ×

Select destination server

DESTINATION SERVER
WIN2019-SERVER

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Confirmation

Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool

☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
WIN2019-SERVER	169.254.236.128	Microsoft Windows Server 2019 Standard Evaluation

1 Computer(s) found

This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

< Previous

Next >

Install

Cancel

Add Roles and Features Wizard

— □ ×

Confirm installation selections

DESTINATION SERVER
WIN2019-SERVER

Before You Begin

Installation Type

Server Selection

Server Roles

Features

AD DS

Confirmation

Results

To install the following roles, role services, or features on selected server, click Install.

☒ Restart the destination server automatically if required

Optional features (such as administration tools) might be displayed on this page because they have been selected automatically. If you do not want to install these optional features, click Previous to clear their check boxes.

Active Directory Domain Services

Group Policy Management

Remote Server Administration Tools

Role Administration Tools

AD DS and AD LDS Tools

Active Directory module for Windows PowerShell

AD DS Tools

Active Directory Administrative Center

AD DS Snap-Ins and Command-Line Tools

Export configuration settings

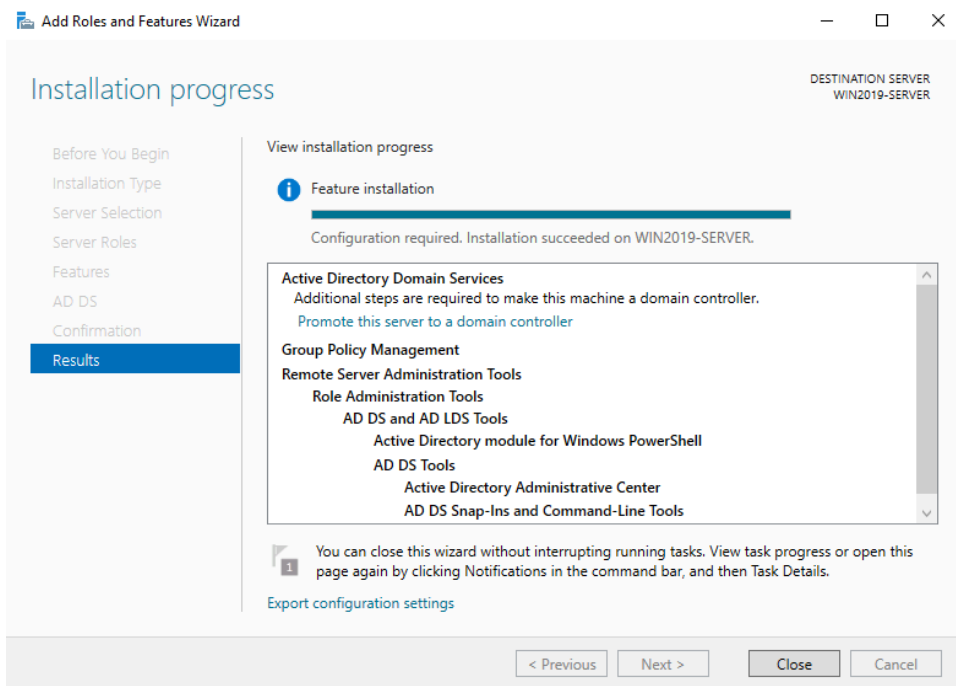
Specify an alternate source path

< Previous

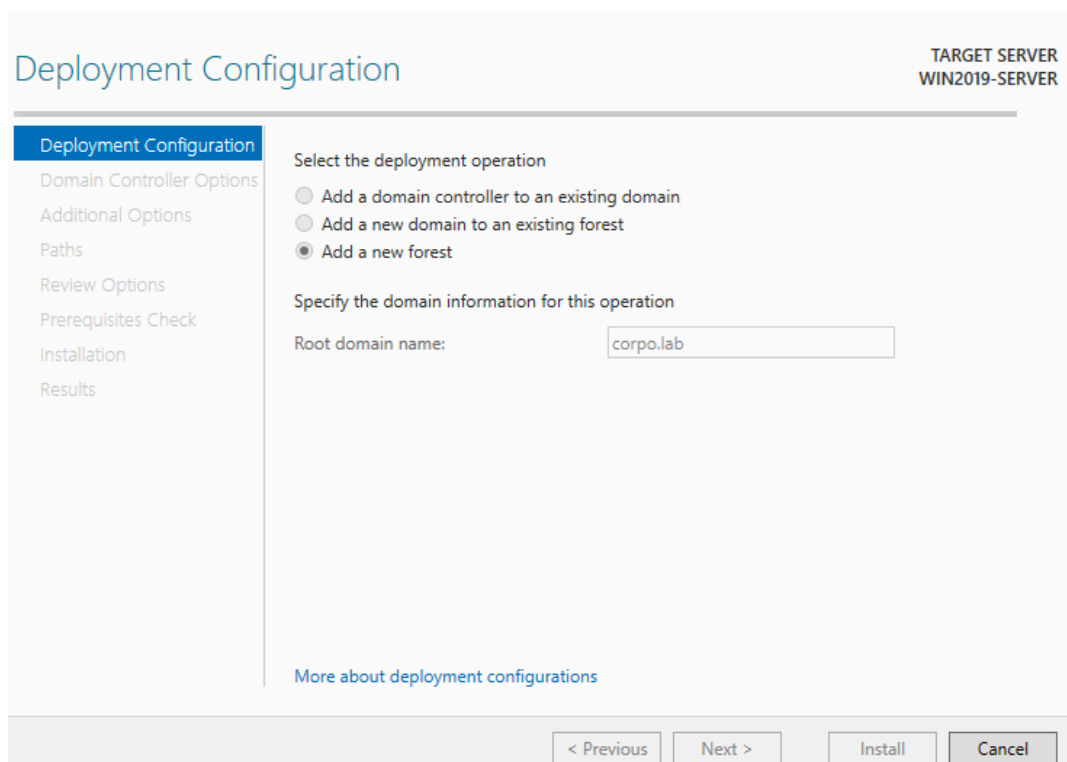
Next >

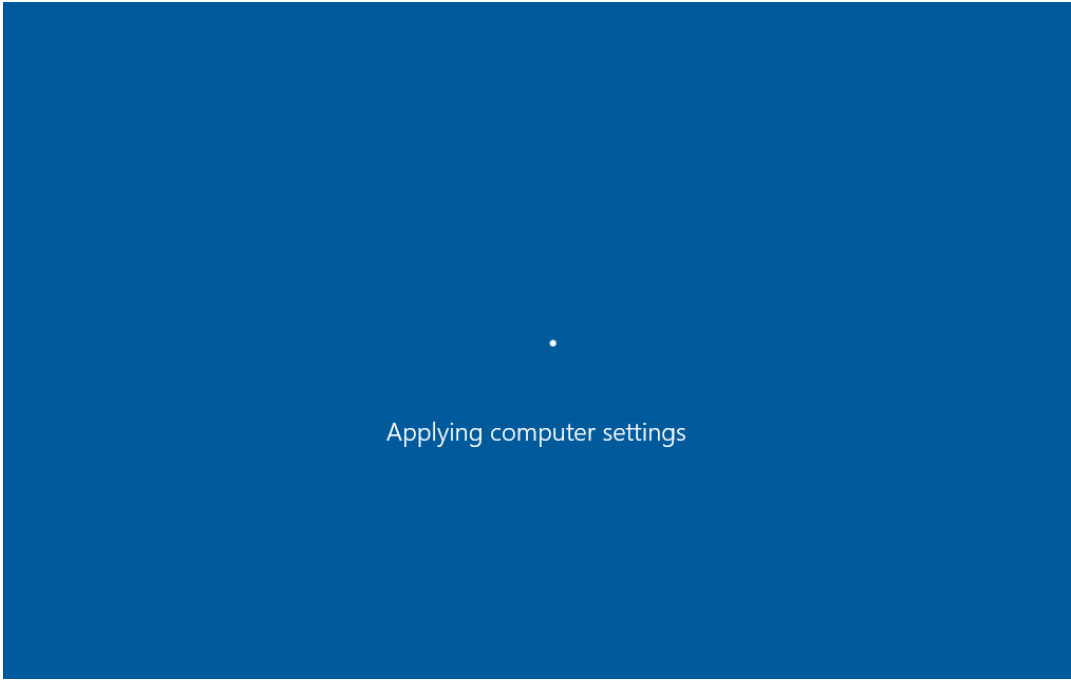
Install

Cancel

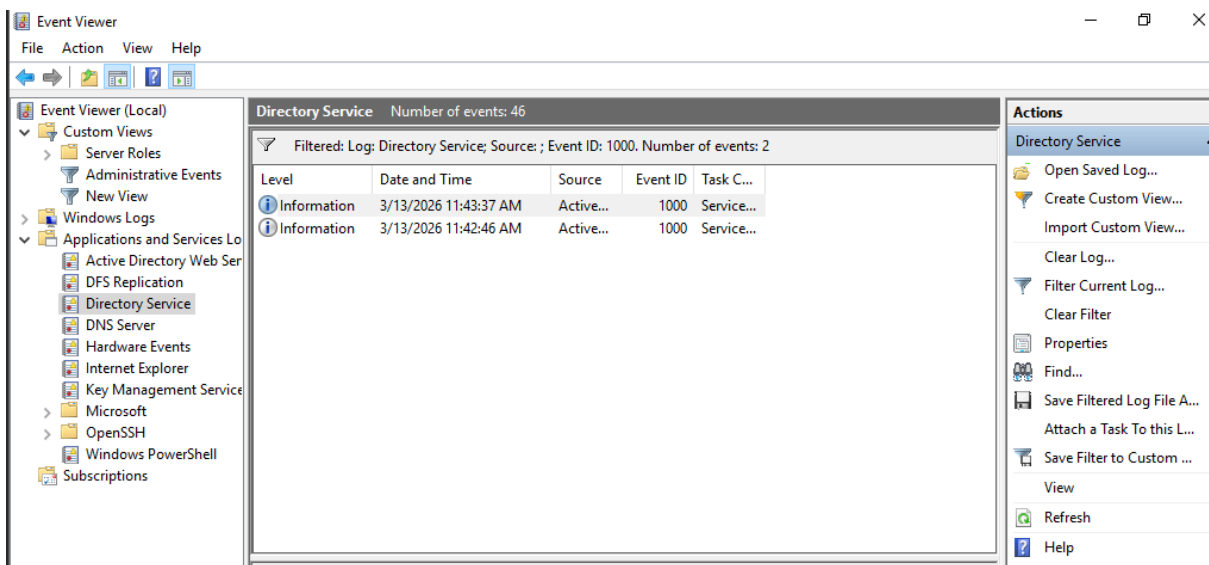
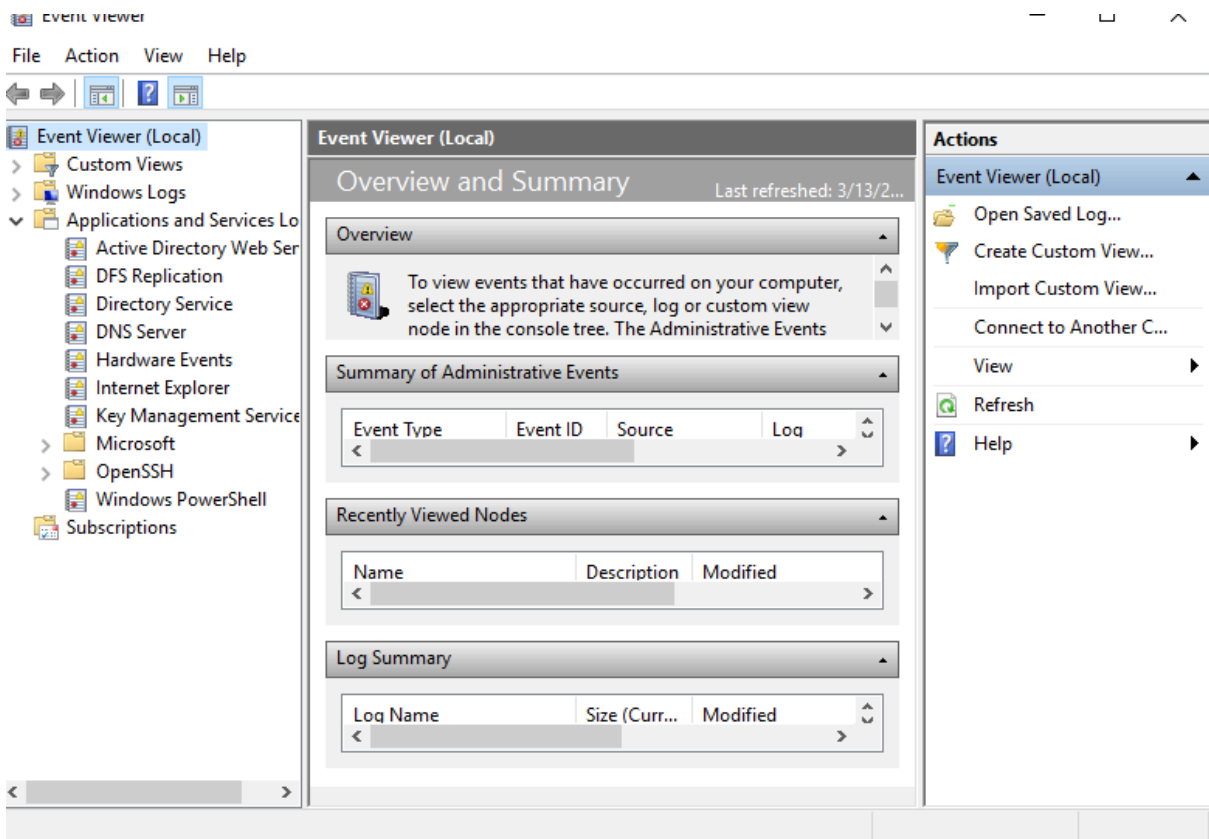


4a-4j

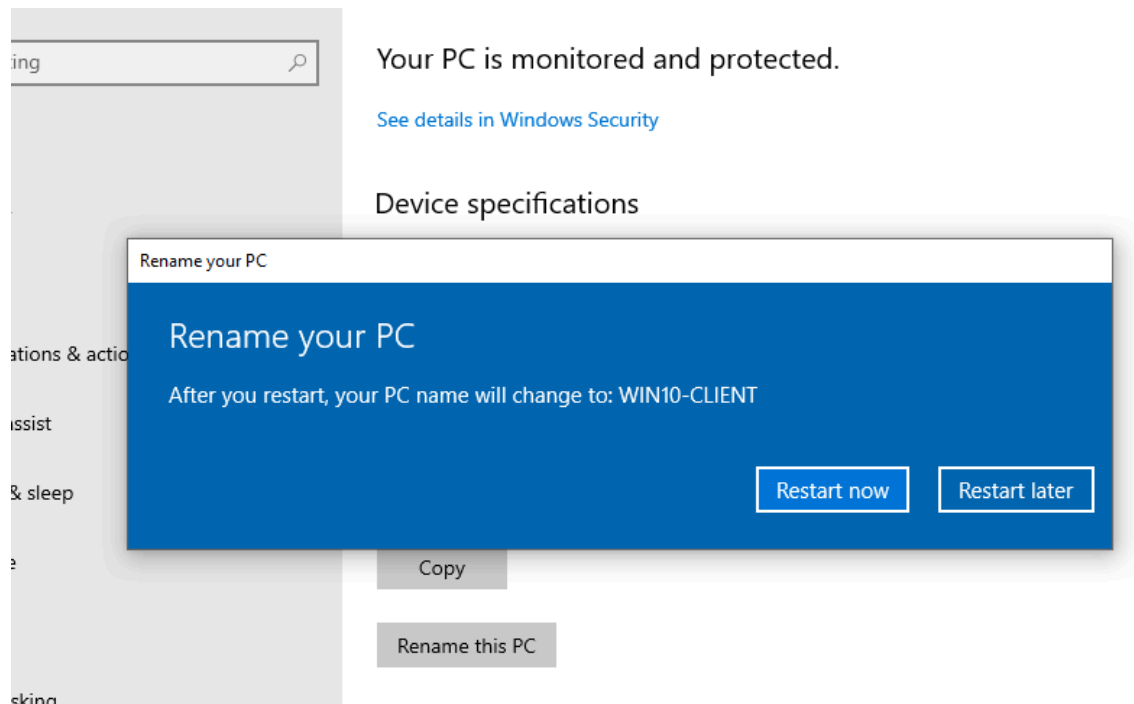




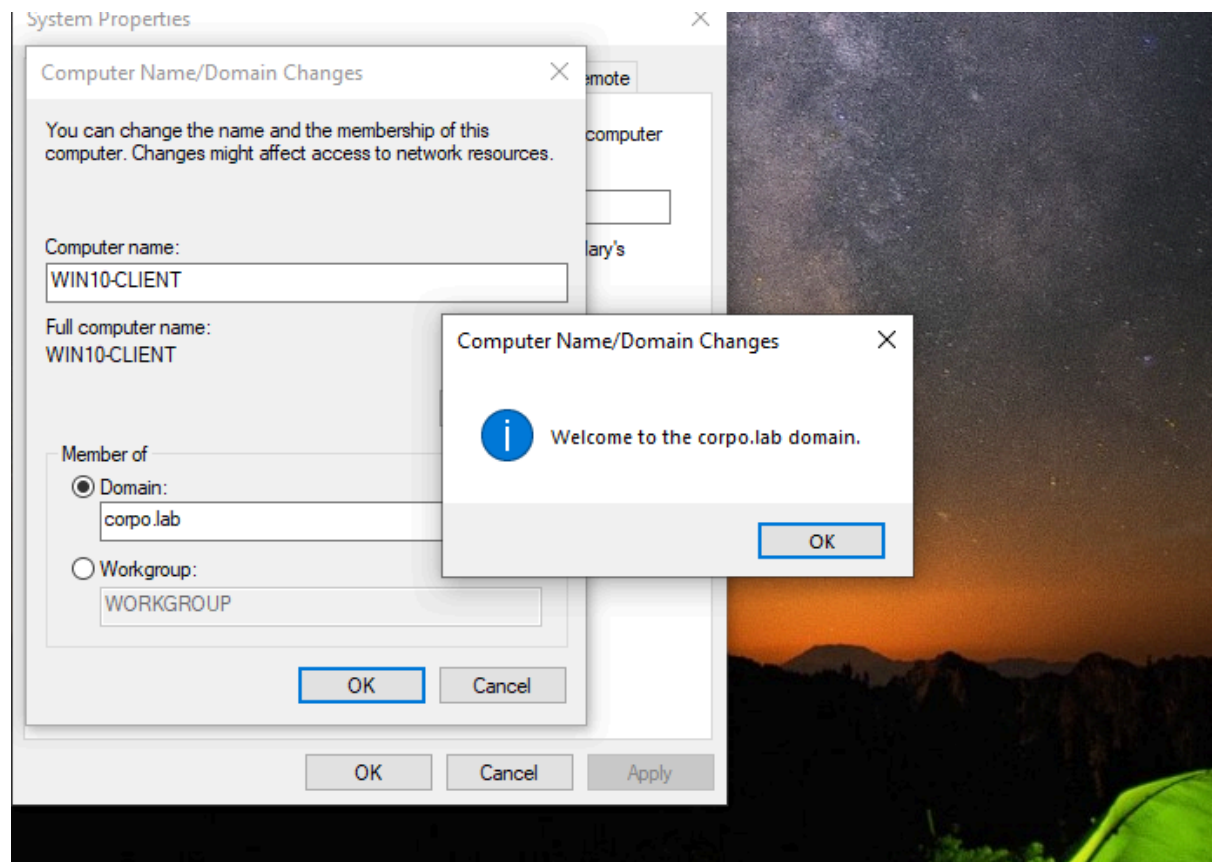
5a-5e



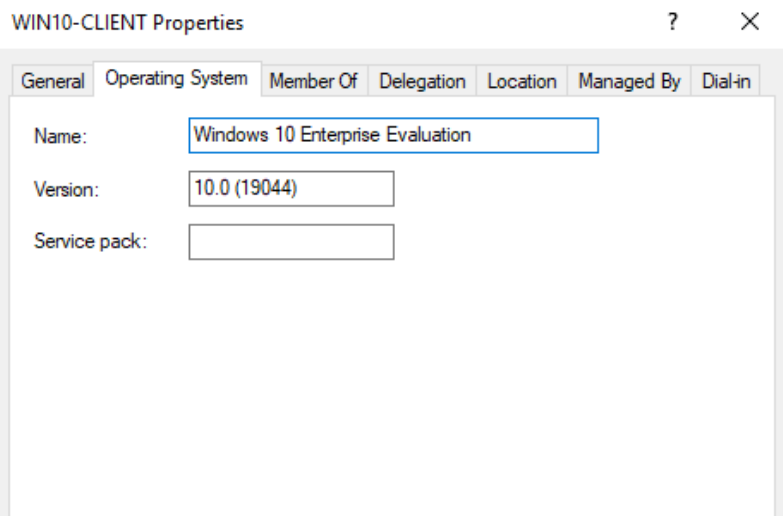
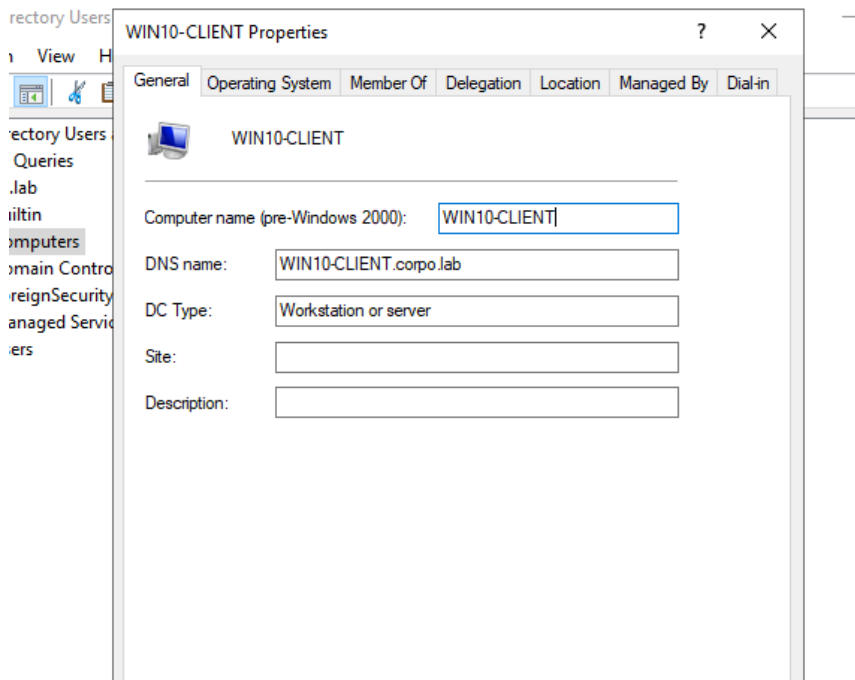
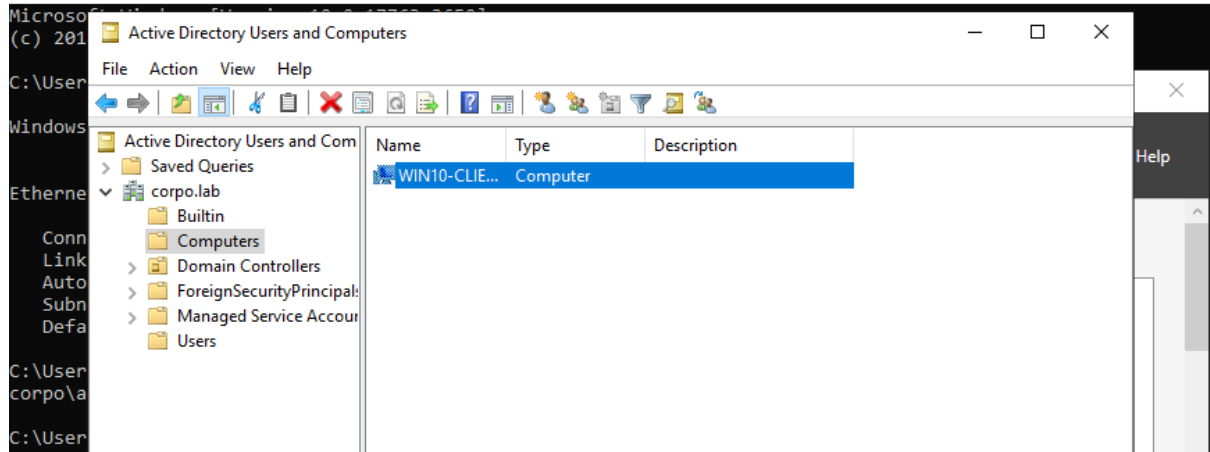
6



7a-7f

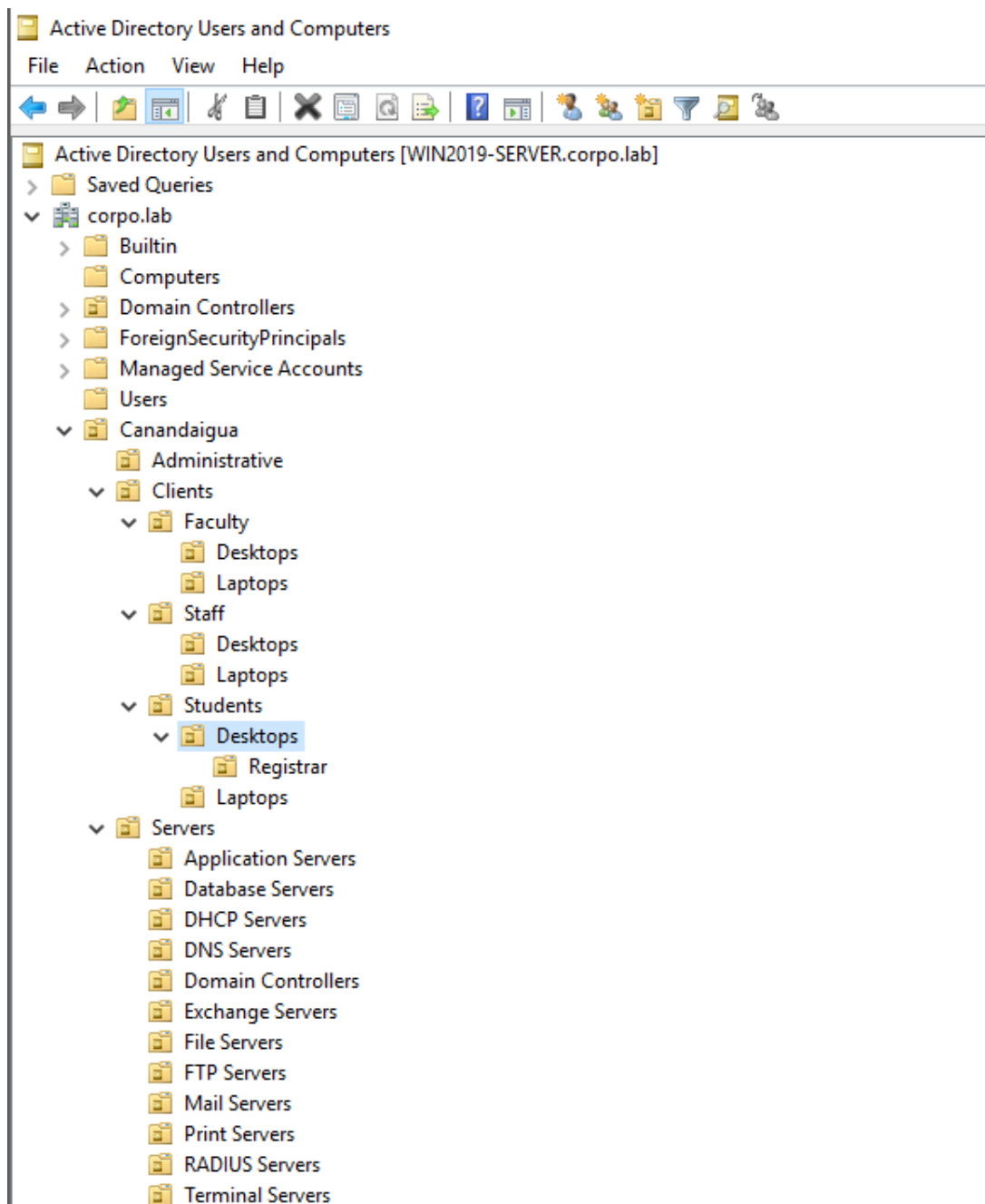


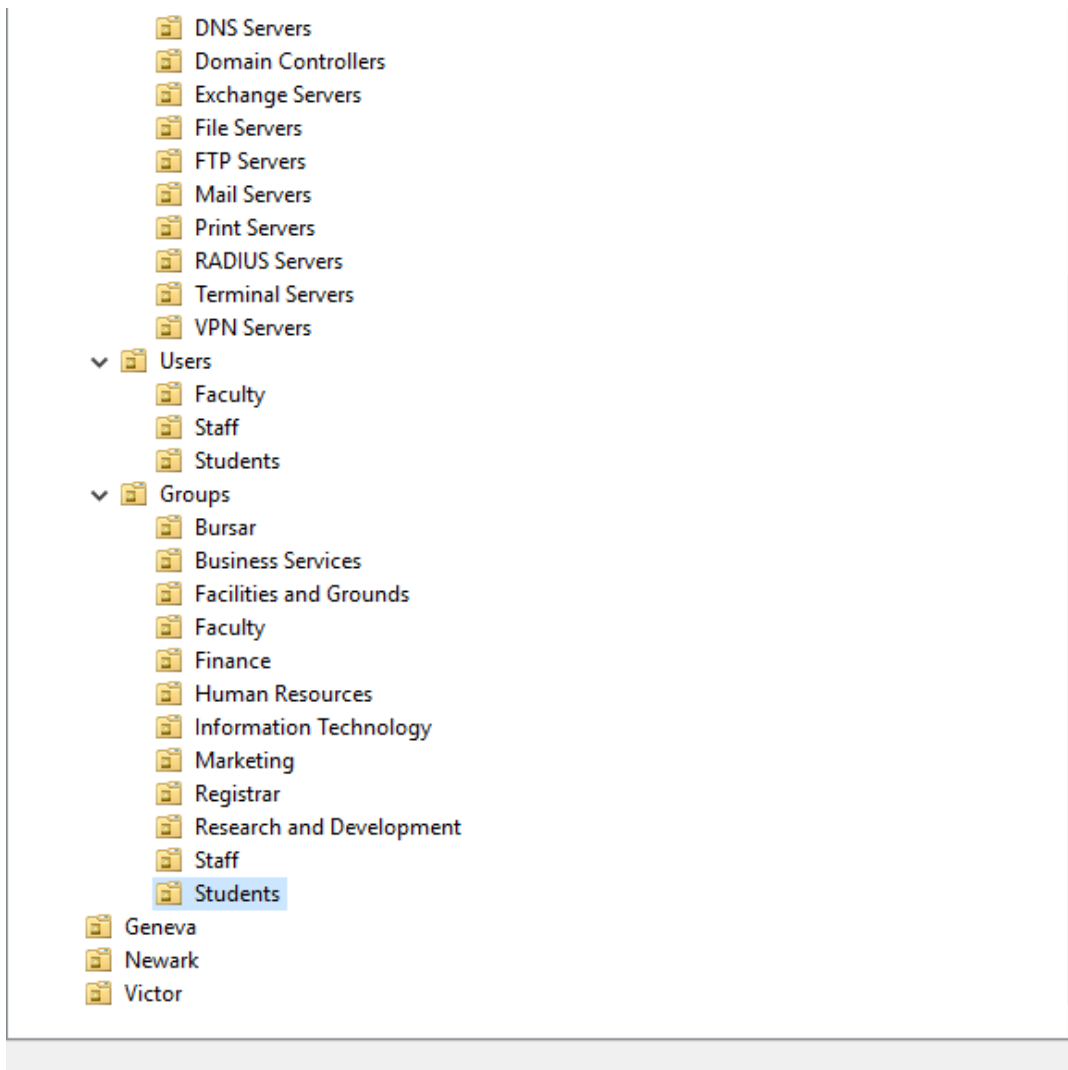
Administrator: Command Prompt



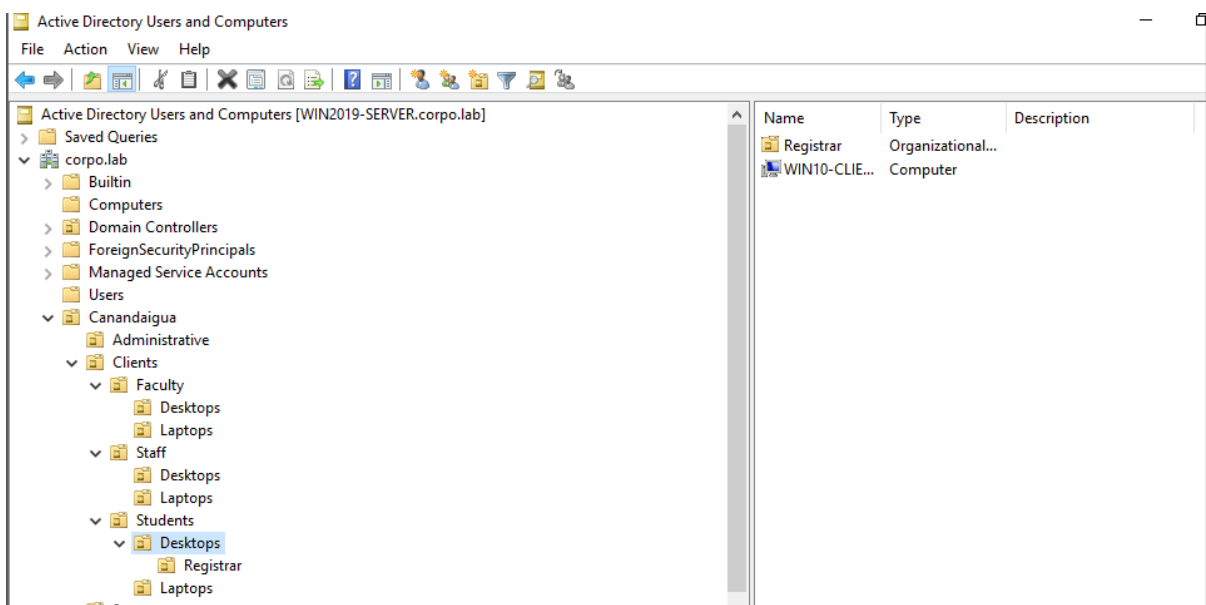
Lab Exercise 14.02: Organizational Units and Groups

1a-1k





2a-2f



3a-3i

Delegation of Control Wizard

Select Users, Computers, or Groups

Select Users, Computers, or Groups

Select this object type:
Users, Groups, or Built-in security principals

Object Types...

From this location:
corpo.lab

Locations...

Common Queries

Name: Starts with

Description: Starts with

☐ Disabled accounts

☐ Non expiring password

Days since last logon:

Columns...

Find Now

Stop

Search results:

Name	E-Mail Address	Description	In Folder
Access Control Assistance Operators			corpo.lab/E
Account Operators			corpo.lab/E
Administrator		Built-in account f...	corpo.lab/U
Administrators			corpo.lab/E
Allowed RODC Password Replication Group		Members in this ...	corpo.lab/U
ANONYMOUS LOGON			
Authenticated Users			
Authentication authority asserted identity			
Backup Operators			corpo.lab/E

OK Cancel

Account Operators Properties

General Members Member Of Managed By

Account Operators

Group name (pre-Windows 2000): Account Operators

Description: Members can administer domain user and group account

E-mail:

Group scope

☒ Builtin local

☐ Global

☐ Universal

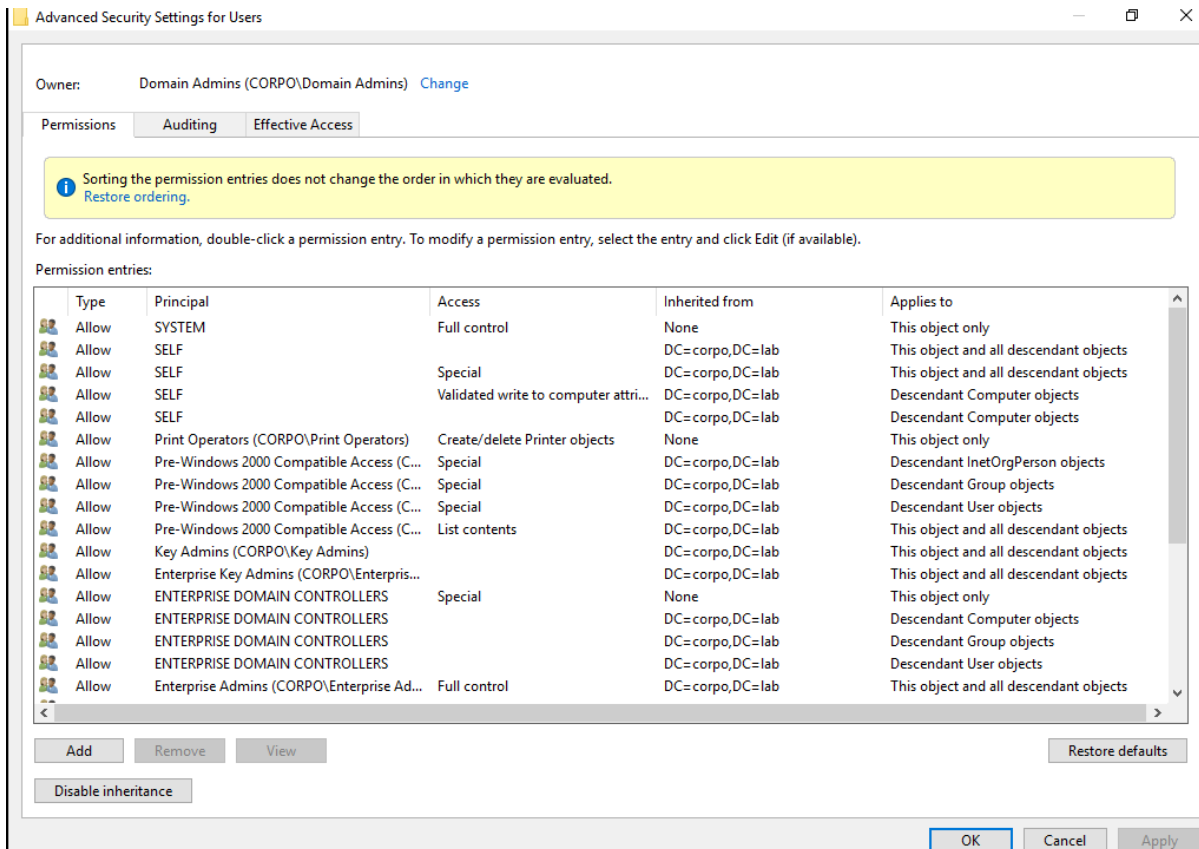
Group type

☒ Security

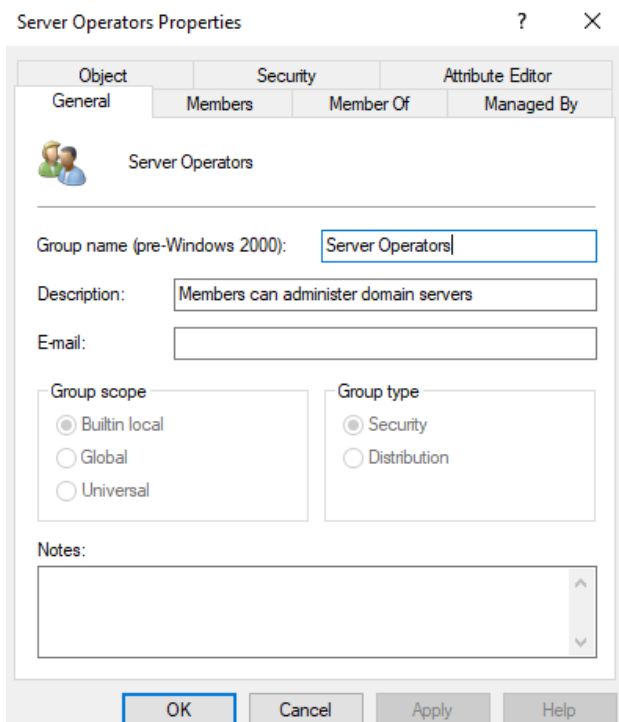
☐ Distribution

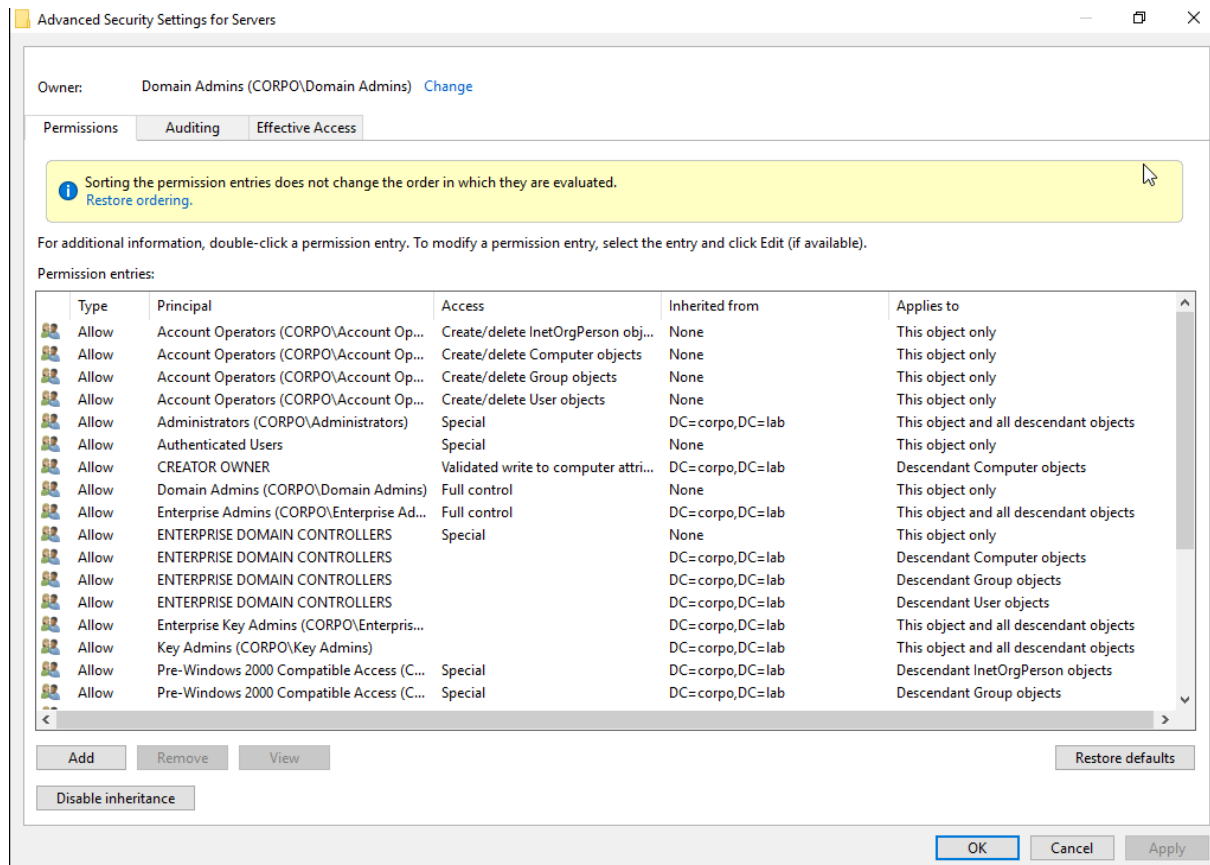
Notes:

OK Cancel Apply

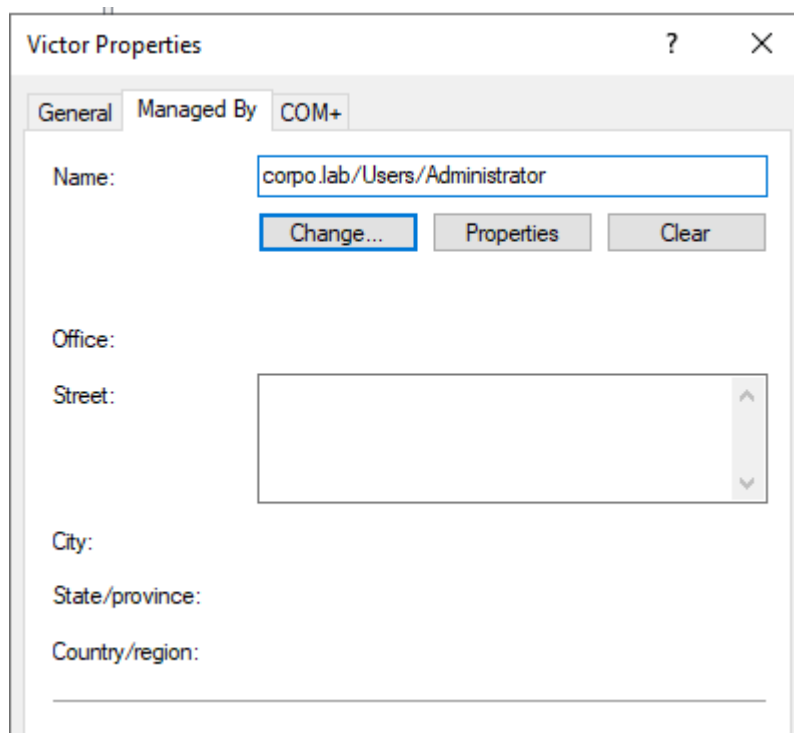


4a-4n





5a-5c



6a-6c

Active Directory Users and Computers

File Action View Help

Active Directory Users and Computers [WIN2019-SERVER.corpo.lab]

Name	Type	Description
HR Managers	Security Group...	

Navigation pane:

- Active Directory Users and Computers [WIN2019-SERVER.corpo.lab]
 - Saved Queries
 - corpο.lab
 - Builtin
 - Canandaigua
 - Administrative
 - Clients
 - Groups
 - Business Services
 - Facilities and Grounds
 - Faculty
 - Finance
 - Human Resources

Active Directory Users and Computers

File Action View Help

Active Directory Users and Computers [WIN2019-SERVER.corpo.lab]

Name	Type	Description
IT Level 1	Security Group...	

Navigation pane:

- Active Directory Users and Computers [WIN2019-SERVER.corpo.lab]
 - Saved Queries
 - corpο.lab
 - Builtin
 - Canandaigua
 - Administrative
 - Clients
 - Groups
 - Business Services
 - Facilities and Grounds
 - Faculty
 - Finance
 - Human Resources
 - Information Technology
 - Marketing

Active Directory Users and Computers [WIN2019-SERVER.corpo.lab]

Name	Type	Description
Honors Stud...	Security Group...	

Navigation pane:

- Active Directory Users and Computers [WIN2019-SERVER.corpo.lab]
 - Saved Queries
 - corpο.lab
 - Builtin
 - Canandaigua
 - Administrative
 - Clients
 - Groups
 - Business Services
 - Facilities and Grounds
 - Faculty
 - Finance
 - Human Resources
 - Information Technology
 - Marketing
 - Research and Development
 - Staff
 - Students

Lab Exercise 14.03: Users and Other Active Directory Objects

1a-1e

New Object - User ✕

 Create in: `corpo.lab/Canandaigua/Users/Staff`

When you click Finish, the following object will be created:

Full name: David L. Goggins

User logon name: davgogi@corpo.lab

< Back Finish Cancel

Active Directory Users and Computers

File Action View Help



Active Directory Users and Computers [WIN2019-SERVER.corpo.lab]

Name	Type	Description
David L. Gog...	User	

Left pane tree structure:

- Active Directory Users and Computers [WIN2019-SERVER.corpo.lab]
 - > Saved Queries
 - > corpo.lab
 - > Builtin
 - > Canandaigua
 - > Administrative
 - > Clients
 - > Groups
 - > Servers
 - > Users
 - > Faculty
 - > Staff
 - > Students

2a-2b

New Object - User

 Create in: `corpo.lab/Canandaigua/Users/Students`

When you click Finish, the following object will be created:

Full name: Luke B. Bourgh

User logon name: luccibog@corpo.lab

The user must change the password at next logon.

< Back Finish Cancel

Active Directory Users and Computers

File Action View Help



Active Directory Users and Computers [WIN2019-SERVER.corpo.lab]

> Saved Queries

▼  `corpo.lab`

> Builtin

▼  Canandaigua

> Administrative

> Clients

> Groups

> Servers

▼  Users

> Faculty

 Staff

 Students

Name	Type	Description
 Luke B. Bour...	User	

3a-3f

David L. Goggins Properties



Security	Environment	Sessions	Remote control
Remote Desktop Services Profile		COM+	Attribute Editor
General	Address	Account	Profile
Published Certificates	Member Of	Telephones	Organization
	Password Replication	Dial-in	Object

Member of:

Name	
Domain Admins	Active Directory Domain Services Folder
Domain Users	corpo.lab/Users

Add... Remove

Administrators Properties



General	Members	Member Of	Managed By
Object	Security	Attribute Editor	

Canonical name of object:

corpo.lab/Builtin/Administrators

Object class: Group

Created: 3/13/2026 11:42:46 AM

Modified: 3/13/2026 11:58:37 AM

Update Sequence Numbers (USNs):

Current: 12752

Original: 8199

☐ Protect object from accidental deletion

OK Cancel Apply Help

Administrators Properties ? X

Object		Security		Attribute Editor	
General		Members		Member Of	
Members:					
Name	Active Directory Domain Services Folder				
 Administrator	corpo.lab/Users				
 Domain Admins	corpo.lab/Users				
 Enterprise Ad...	corpo.lab/Users				

Add... Remove

OK Cancel Apply Help

Domain Admins Properties ? X

Object		Security		Attribute Editor	
General		Members		Member Of	
 Domain Admins					
Group name (pre-Windows 2000): Domain Admins					
Description: Designated administrators of the domain					
E-mail:					
Group scope			Group type		
☐ Domain local			☒ Security		
☒ Global			☐ Distribution		
☐ Universal					
Notes:					

OK Cancel Apply Help

Honors Students Properties ? X

Object		Security	Attribute Editor	
General	Members	Member Of	Managed By	
Members:				
Name	Active Directory Domain Services Folder			
 David L. Gog...	corpo.lab/Canandaigua/Users/Staff			
Add... Remove				
OK Cancel Apply Help				

4a-4c

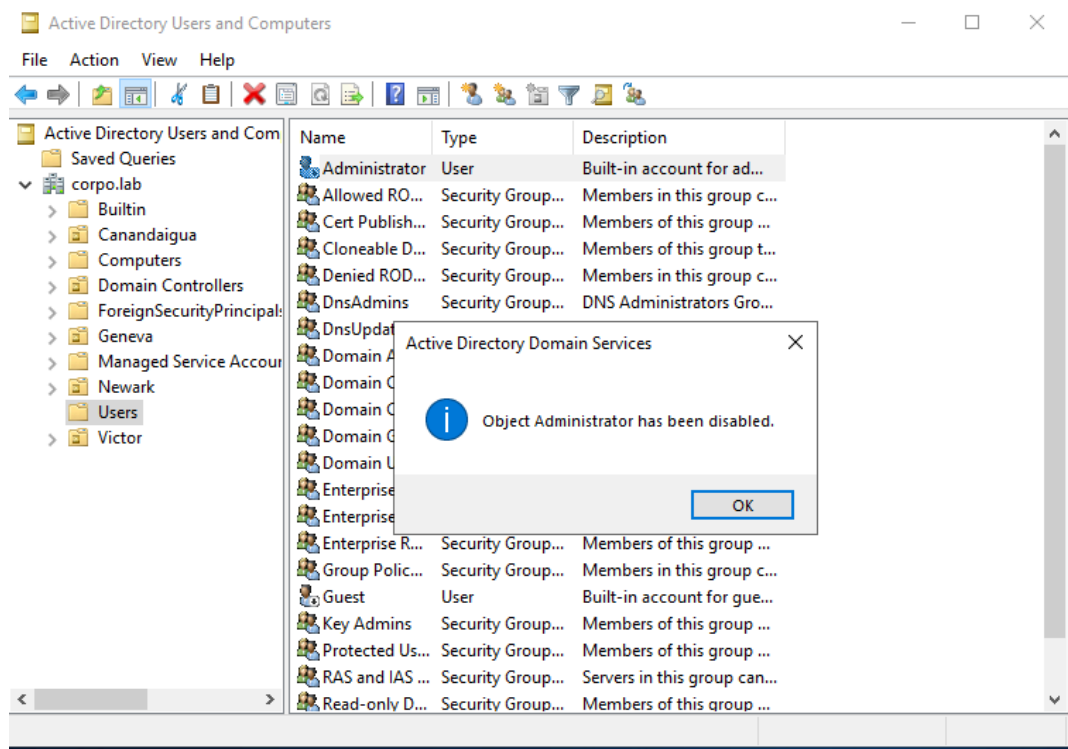
Luke B. Bourgh Properties ? X

Published Certificates	Member Of	Password Replication	Dial-in	Object
Security	Environment	Sessions	Remote control	
Remote Desktop Services Profile		COM+	Attribute Editor	
General	Address	Account	Profile	Telephones
 Luke B. Bourgh				
First name: Luke Initials: B				
Last name: Bourgh				
Display name: Luke B. Bourgh				
Description: good guy				
Office: Shanghai, China				
Telephone number: 0974334163 Other...				
E-mail: d23401dvv@gmail.com				
Web page: http://luke.com Other...				
OK Cancel Apply Help				

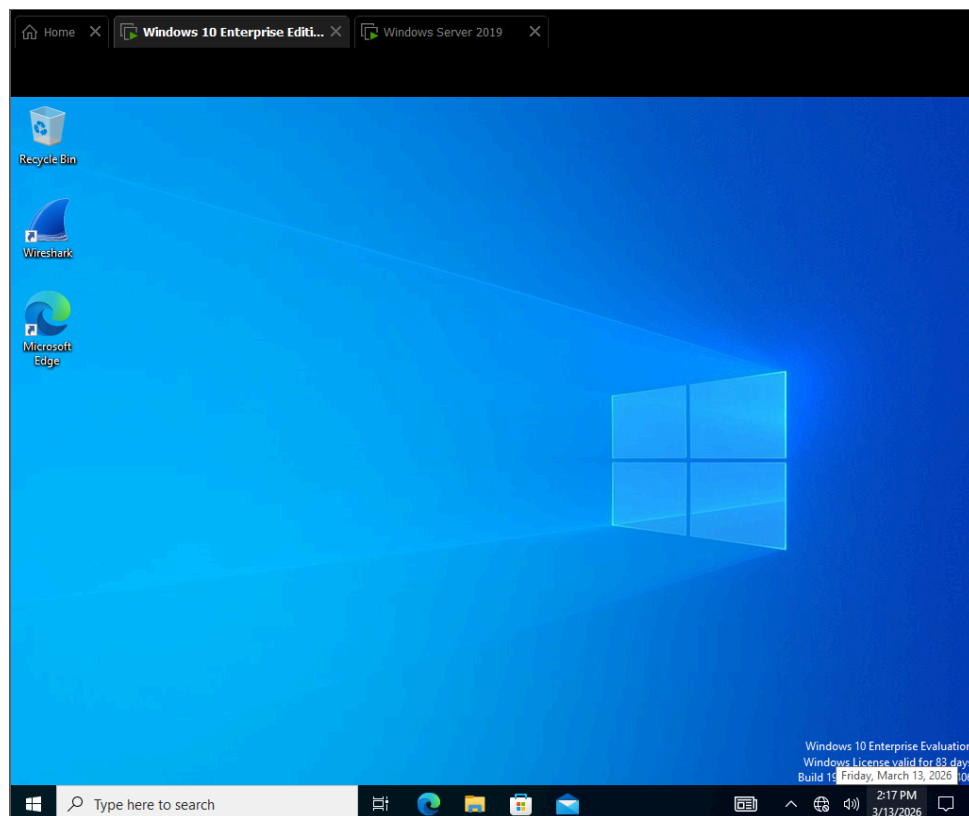
Luke B. Bourgh Properties ? X

Published Certificates	Member Of	Password Replication	Dial-in	Object
Security	Environment	Sessions	Remote control	
Remote Desktop Services Profile		COM+	Attribute Editor	
General	Address	Account	Profile	Telephones
Street: fewfewfewf				
P.O. Box: 134				
City: Shanghai				
State/province: Zhinghia				
Zip/Postal Code: 1244				
Country/region: China				
OK Cancel Apply Help				

5a-5e



-Win10 VM sign in from new non-admin account:



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

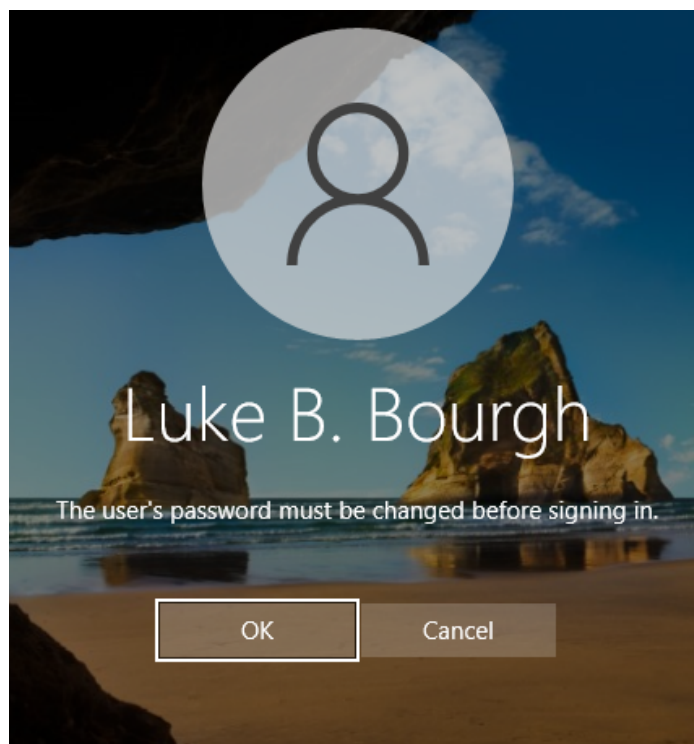
PS C:\Users\luccibog> whoami
corp\luccibog
PS C:\Users\luccibog>
```

6a-6b

The screenshot shows the 'Active Directory Users and Computers' console window. The left pane displays the tree structure with 'Users' expanded. The right pane shows a table with one user entry:

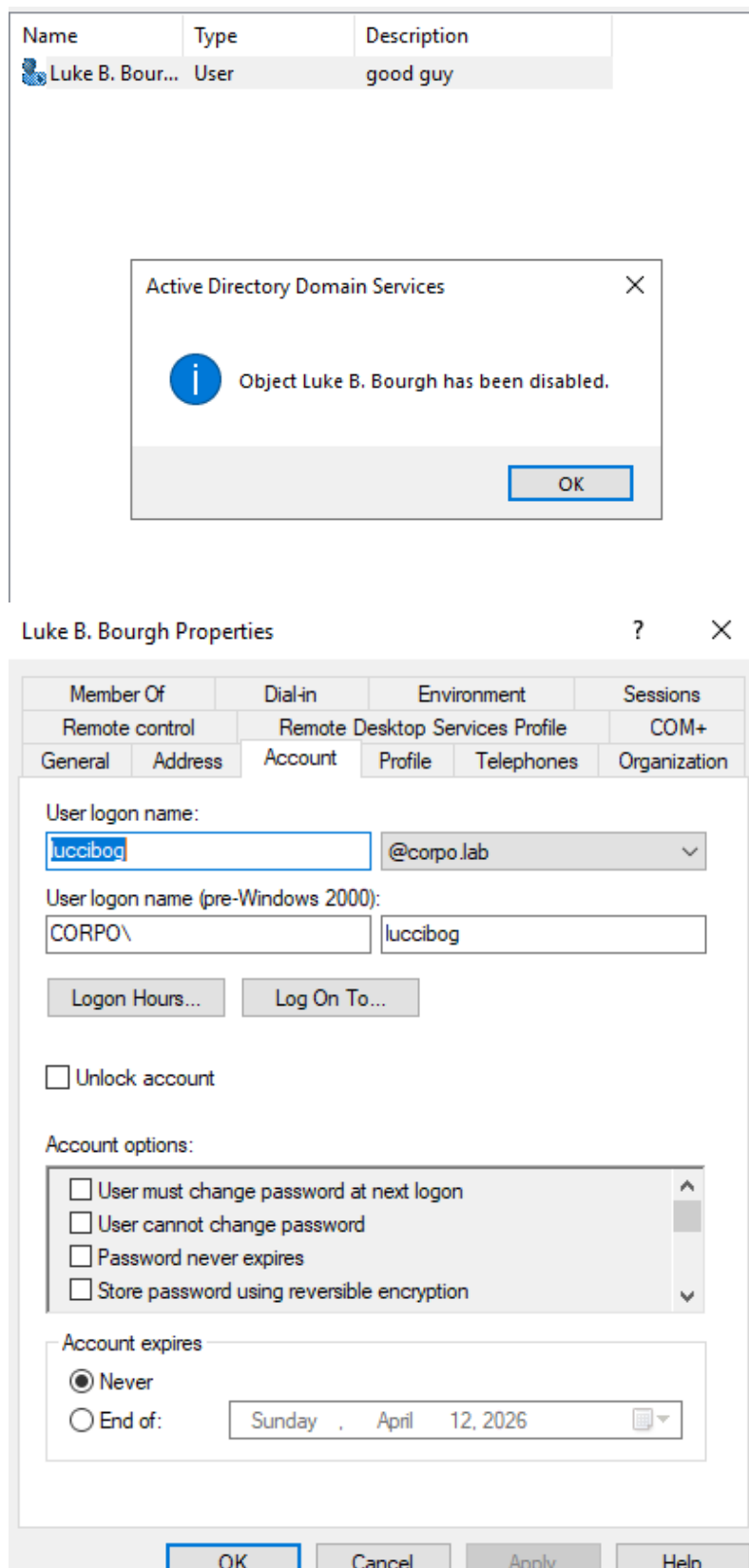
Name	Type	Description
Luke B. Bour...	User	good guy

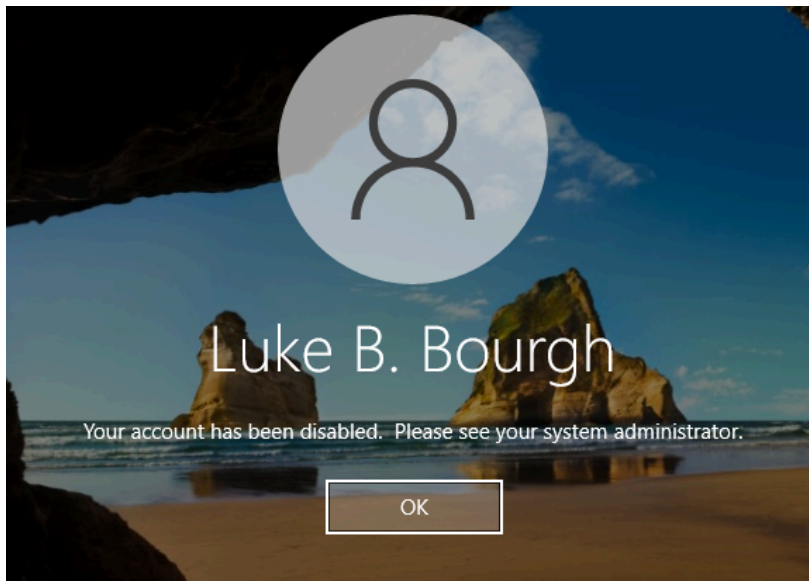
Overlaid on the console is a small 'Active Directory Domain Services' dialog box with an information icon and the message: 'The password for Luke B. Bourgh has been changed.' with an 'OK' button.



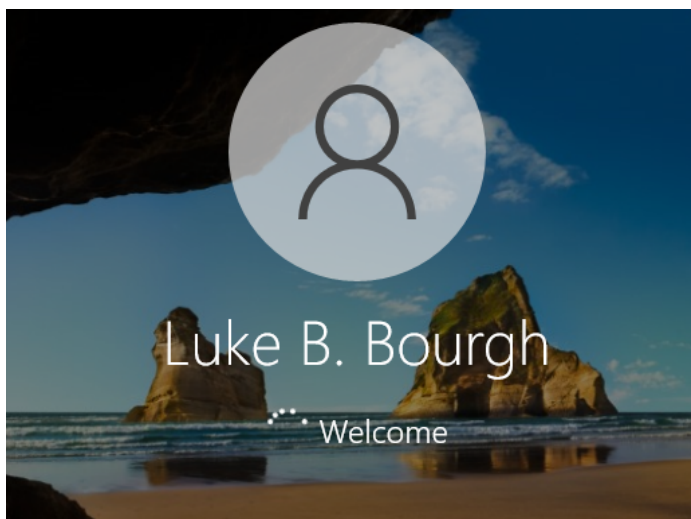
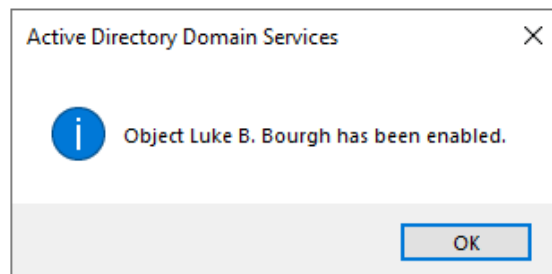
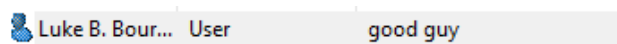
7a-7g

-Disable the non-admin account from Win Server 2019:



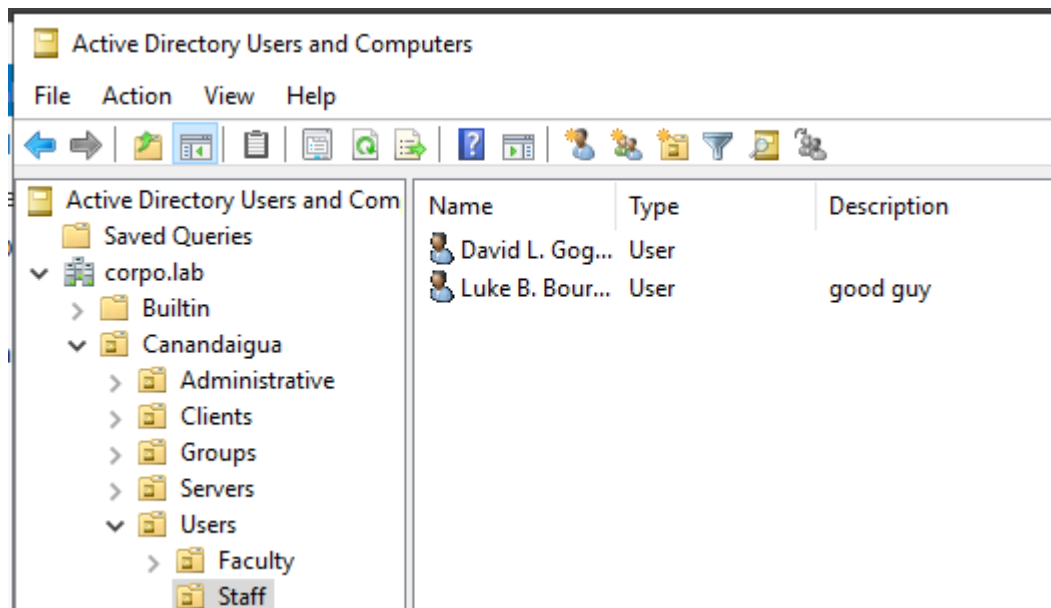


-Re-enable the non-admin account:

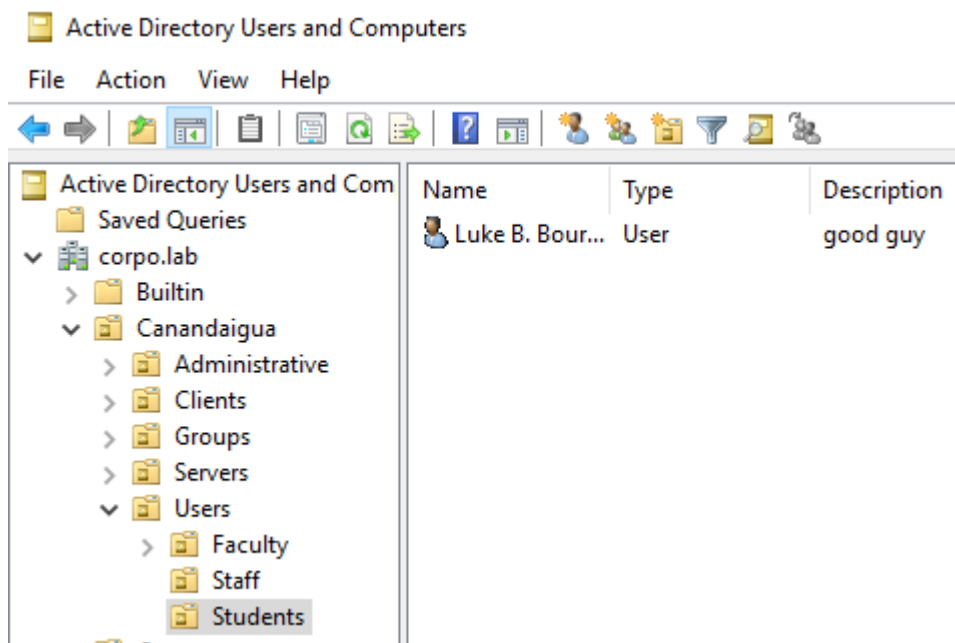


8a-8d

-Move non-admin user to Staff OU:



-Move non-admin user back to Students OU:



Rename User ? X

Full name: Luke B. Bander

First name: Luke

Last name: Bander

Display name: Luke B. Bander

User logon name:
luccibog @corpo.lab

User logon name (pre-Windows 2000):
CORPO\ b\uccibog

OK Cancel

-Create new user in Students OU:

New Object - User X

Create in: corpo.lab/Canandaigua/Users/Students

First name: Grace Initials: S

Last name: Shills

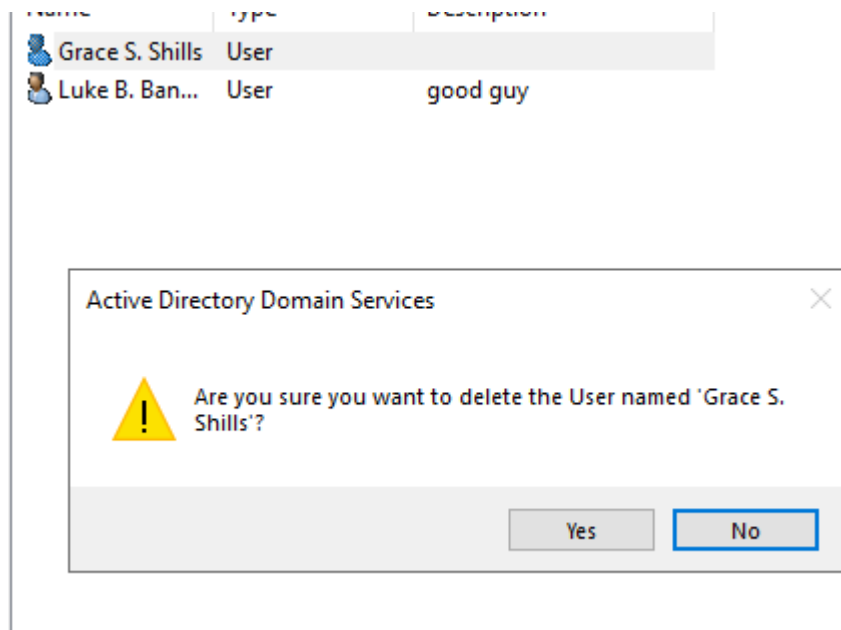
Full name: Grace S. Shills

User logon name:
grashills @corpo.lab

User logon name (pre-Windows 2000):
CORPO\ grashills

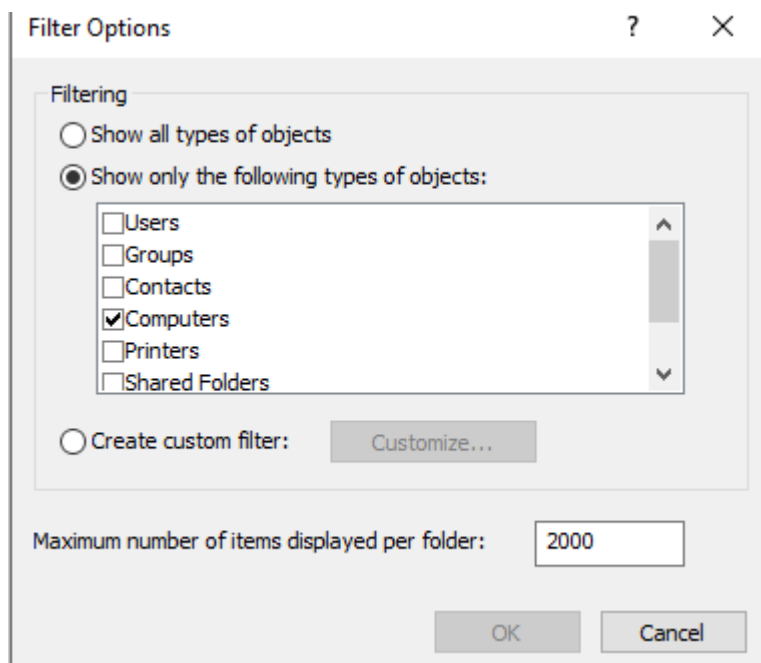
< Back Next > Cancel

-Prompt to delete the newly created user:



9a-9b

-Apply filter for view (Only Computers Object are visible):



Active Directory Users and Computers

FileActionViewHelp

Active Directory Users and Computers

corp0.lab

Built-in

Canandaigua

Administrative

Clients

Faculty

Staff

Students

Desktops

Laptops

Name	Type	Description
Registrar	Organizational...	
WIN10-CLIE...	Computer	

10a-10g

 Find Users, Contacts, and Groups

File Edit View

Find: Users, Contacts, and Groups In:  corpo.lab Browse...

Users, Contacts, and Groups Advanced

Name: Domain

Description:

Find Now Stop Clear All



Search results:

Name	Type	Description
 Domain Users	Group	All domain users
 Domain Guests	Group	All domain guests
 Domain Control...	Group	All domain controllers in the doma
 Domain Compu...	Group	All workstations and servers joined
 Domain Admins	Group	Designated administrators of the dc

< 5 item(s) found >

-Find existing computers inside domain with dropdown Computers:

Find: Computers In: corpo.lab Browse...

Computers Advanced

Computer name:

Owner:

Role: Any

Find Now Stop Clear All

Search results:

Name	Machine Role	Owner
WIN10-CLIENT	Workstation or Server	
WIN2019-SERVER	Writable Domain Controller	

2 item(s) found

-Find existing user with last name that starts with the letter “b” from the domain database:

Find: Users, Contacts, and Groups In: corpo.lab Browse...

Users, Contacts, and Groups Advanced

Field Condition: Value:

Condition List: Add Remove

Last Name Starts with b

Find Now Stop Clear All

Search results:

Name	Type	Description
Luke B. Bander	User	good guy

1 item(s) found